ATOMIC: an Interpretable Clustering Method Based on Data Topology

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ABSTRACT

State-of-the-art clustering algorithms are well equipped to partition a dataset, but provide no insight into their process for selecting a particular partition. As such clustering strategies which work in an interpretable manner, i.e., in a way that can be understood, are desired. Current explainable clustering approaches focus mainly on explaining k-means clustering, which limits their scope to problems where clusters are spherical and convex. Additionally, many of these algorithms struggle to produce explanations on high dimensional data due to their computational complexity. We propose an interpretable clustering methodology which addresses these challenges. Our algorithm, ATOMIC: Analysis of Topology Oriented Method for Interpretable Clustering, is designed to answer the question "why does my data fit into distinct clusters?". This is done by identifying a set of variables that are responsible for isolating a cluster of data-points from the rest in the dataset. To locate partitions which are defined by specific variables we utilize Novelty Search with Local Competition. ATOMIC relies on basic concepts from topology to partition the dataset. We test our algorithm on well-studied high-dimensional datasets, along with performance comparisons to state-of-the-art clustering methodologies. We compare our algorithm in terms of interpretability to other interpretable and explainable clustering methodologies.

CCS CONCEPTS

• **Computing methodologies** → **Genetic algorithms**; **Cluster analysis**; • **Mathematics of computing** → Algebraic topology.

KEYWORDS

Interpretable Clustering, Clustering Methodology, Explainable Artificial Intelligence

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1 INTRODUCTION

There are two main challenges facing current state-of-the-art explainable and interpretable clustering methods. The first is the vast majority of them are designed to explain k-means clustering [9, 10, 20]. This classic clustering algorithm is effective, however it works best to detect convex, non-overlapping clusters. The second challenge is scaling these methods into higher dimensional settings. Many algorithms are based on threshold trees and scale poorly with the number of features in the data. In our experimentation it was challenging to find algorithms which could cluster data with over 10 dimensions in an interpretable manner. We present a novel algorithm ATOMIC: Analysis of Topology Oriented Method for Interpretable Clustering which addresses these issues. Our algorithm utilizes Novelty Search with Local Competition (NSLC) and basic concepts from topology to cluster data in an interpretable manner. ATOMIC can find a wider variety of cluster shapes than k-means clustering, and is explicitly designed to work on higher dimensional clustering problems.

1.1 Explainable and Interpretable Clustering

State of the art clustering approaches are black boxes, meaning they "provide little insight into the rationale for cluster membership" [3]. The lack of clarity from these methods has lead to a boom in research on explainable and interpretable clustering [5]. Current research categorizes explanation techniques principally into two groups, explainability and interpretability [21]. We adopt the particular distinction presented by Rudin [17] where explainable techniques exist externally from and run after a main black-box model. Since post-hoc explanatory techniques must be simpler than the models they explain there is an approximation factor between black-box models and the associated explainability techniques [6].

We set out to avoid having to deal with the approximation issues of explainable approaches, hence we build an interpretable model. Interpretability as defined by Murdoch is "the extraction of relevant knowledge from a machine-learning model concerning relationships either contained in data or learned by the model" [14]. We suggest that information which answers the question *Why does my data fit into distinct clusters?* will be precisely the "relevant knowledge" mentioned by Murdoch. Using an interpretable model means we can confidently assert which variables are responsible for our clustering result without fear of this being a product of an error in the post-hoc model.

2 METHODOLOGY

We are inspired by Naitzat's use of topology to track how a dataset is split into distinct groups as it is processed by a neural network [15]. While that paper seeks to track the changes in the topology of a

Table 1: Summary of Parameters and Methods used in Evolutionary Algorithm

Representation	Boolean List of length p
Mutation	Flip random entry
Mutation Probability	60%
Crossover	Bit-wise OR
Crossover probability	40%
Survivor Selection	NSGA-II selection
Population Size	100
Number of Offspring	60
Initialisation	Random
Objective	Sparseness and Local Competition
Termination Condition	Reached 100 generations

dataset as it is passed through a neural network to find a split, we wish to flip this process on its head. We desire to find a topological space for our data which creates a split. We seek to find a minimal subset of variables which is sufficient to isolate a group of data-points from the rest of the dataset. If we view the isolated group as a cluster, we then have a set of variables which are responsible for the formation of that cluster. This approach links variables explicitly to a specific cluster, whereas feature selection simply finds a set of variables which correspond to a clustering algorithm performing well. Even if we limit a clustering algorithm to two clusters, the variables found by feature selection are what optimize a particular cluster quality metric, not what is responsible for creating the distinction between a cluster and the rest of the dataset. In our approach we explicitly find the variables responsible for isolating data-points; this is the exact answer to our interpretability test of "Why does my data fit into distinct clusters?" This approach in addition to providing interpretability, helps locate non-convex and overlapping clusters. By looking through multiple topological spaces, we can find isolated groups which in the original topological space are not neatly distinguished. Pairing this approach with an evolutionary search allows us to apply this to problems of high dimensionality.

2.1 Topology Warping

We begin by noting that when *variables* are used by a clustering algorithm we call them *features*. We make this distinction because we will not always cluster using every variable in the dataset. Individuals in our EA are subsets of variables. We wish to test each individual for how well it isolates a cluster of data-points from the rest. We adopt the terminology that a set of variables *isolates a cluster* if using them as features separates the data topology into two connected components. This is similar to the idea of strongly connected component clustering [19]; however, we are explicitly tying the variables to the resulting clusters. We do this because using different variable combinations, which warps the feature space, will lead to different clusters being broken off from the dataset. This phenomena is at the core of ATOMIC's design.

Since we are computing connected components across different feature spaces, each of which will warp a common distance metric, we use a fixed number of nearest neighbours $\tilde{k}$ for determining what data-points are connected in any feature space. The value for $\tilde{k}$ is found using a procedure inspired from [15]. We begin by building a k-nearest neighbours (k-nn) graph of all data-points using Euclidean distance with all original variables of the dataset. We then calculate the number of connected components of this graph. We do this for a range of k values starting from 2, until we reach a stable k. A stable value of k is defined as a value for which there is no change in the number of connected components if k is increased by 1. The stable k is denoted by $\tilde{k}$. With this principle ATOMIC can find the topological connected components in any given feature space.

2.2 ATOMIC Fitness Function

For the sake of interpretability, we desire small sets of variables which isolate a cluster. Similarly we would like our set of variables to define a single partition, i.e define one cluster separated from the rest of the data. This is equivalent to the dataset having two connected components. To capture both goals we need two functions to measure fitness, the size of the feature subset to be minimized and $f(\text{feature set}) = -1 \times (\#\text{ of connected components} - 2)^2$ which is maximized for two connected components.

2.3 NSLC in ATOMIC

We want to find as many well defined clusters in the dataset as exist, hence we want diversity in the clustering solutions found by our algorithm. Novelty Search proposed by Lehman and Stanley focuses on searching for behavioural novelty [12], which in our case would be the separation of different clusters. Novelty Search with Local Competition (NSLC) works similarly but also allows optimization for small subsets of variables [13]. NSLC defines two objective functions both to maximize, novelty and local competition.

For our novelty metric we use sparseness as suggested by Lehman and Stanley, which is the average distance to the k nearest neighbours of an individual (a set of variables). Our distance metric is the Hamming distance. The local competition metric suggested by the authors is the amount of an individual's k neighbours it outperforms. We define out-performance as follows. Let I_1, I_2 be two individual sets of variables (i.e., individuals in our evolutionary algorithm). Let f be the fitness function. We say that I_1 outperforms I_2 if and only if, $f(I_1) > f(I_2)$ or $(f(I_1) = f(I_2)$ and $|I_1| < |I_2|)$. This ensures that solutions outperform their neighbours only if they either achieve a better clustering result or if they use less features to achieve an equivalent clustering result. This means that our algorithm will search for a diverse set of solutions, but still optimize for better clustering and a smaller amount of defining variables.

In order to calculate both sparsity and local competition we need to know the k nearest neighbours of an individual. To accomplish this we maintain two additional data structures throughout our evolutionary algorithm, both suggested by the authors of NSLC. These are an archive hash-table of previous solutions and a k-nn graph of solutions in the archive. We add solutions to the archive probabilistically with a 0.05% probability for every solution to be added. For the k-nn graph we found $k = 8$ to be a suitable value for k. We summarize the parameters, and methods for our evolutionary algorithm in Table 1.

Table 2: Performance Comparison, using Silhouette metric, between ATOMIC and standard clustering methods

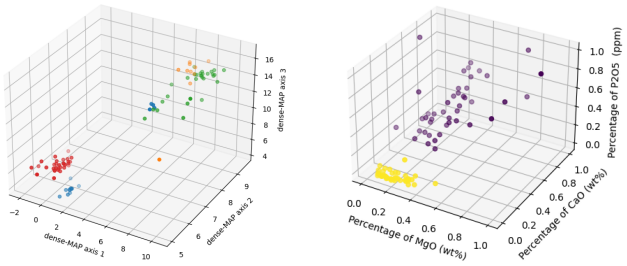
	Silhouette Metric			
Dataset	KMeans	GMM	DBSCAN	ATOMIC
Biodegradation	0.2130	0.0982	0.3275	**0.9132**
Ceramic	0.2129	0.3567	0.1501	**0.6112**
Cervical Cancer	**0.2413**	0.1250	0.0750	0.1118
Glass Identification	0.3233	0.2481	0.3911	**0.5785**
Ionosphere	0.2856	0.1504	0.3386	**0.4004**
Ozone 8-Hours	**0.2135**	0.0344	-0.0890	0.0297
Wholesale	0.6490	0.2187	0.4183	**0.7624**

Table 3: Performance Comparison, using Davies-Bouldin score, between ATOMIC and standard clustering methods

	Davies-Bouldin Score			
Dataset	KMeans	GMM	DBSCAN	ATOMIC
Biodegradation	1.7012	1.8077	1.6995	**0.4457**
Ceramic	1.6626	1.1583	1.8931	**0.5335**
Cervical Cancer	1.5099	**0.9120**	1.8815	0.9272
Glass Identification	1.3810	1.3898	1.5728	**0.4078**
Ionosphere	1.9036	1.7028	2.7599	**0.9106**
Ozone 8-Hours	1.6791	4.4452	1.6459	**1.1201**
Wholesale	0.5997	1.0900	1.2454	**0.3938**

Table 4: Percentage reduction between quantity of variables used by ATOMIC and total number of variables

Dataset	ATOMIC	Total	% Reduction
Biodegradation	2	41	95%
Ceramic	3	18	83%
Cervical Cancer Risk	8	18	55%
Glass Identification	3	9	66%
Ionosphere	10	33	69.9%
Ozone 8-Hours	37	72	48%
Wholesale customers	3	8	62.5%



(a) K-Means Clustering **(b) ATOMIC Clustering**

Figure 1: K-Means versus ATOMIC clustering on Ceramic Dataset

3 RESULTS

3.1 Experimental Setup

We used seven datasets from the UCI Machine Learning Repository [11] for testing ATOMIC's performance. We chose to include datasets with a dimensionality higher than 10 as ATOMIC is designed for clustering high-dimensional data. Before running any clustering methodologies on these datasets, we used basic 2 nearest neighbours imputation to replace missing values, and scaled all values to be within the range $[0, 1]$.

3.2 Cluster Quality Metrics

In order for ATOMIC to be an adequate clustering method, the quality of its resulting solutions must be at least on par with standard clustering algorithms. To determine the quality of clustering solutions we utilize two established metrics, the silhouette score [18] and the Davies–Bouldin index [2]. We use these metrics due to their prominence in clustering literature [8].

3.3 ATOMIC performance comparison

We compare ATOMIC with k-means clustering, Gaussian mixture modelling, and DB-SCAN. We chose these algorithms to cover a wide spread of possible conditions. We chose k-mean clustering since it is the prototypical example of a clustering algorithm and is the most general. We ran this algorithm on a dataset over values of k from 2 to 20 and picked the result which minimized the within-cluster-sum of squared errors, an alternative cluster quality metric. This algorithm performs best at capturing convex non-overlapping clusters. We used Gaussian mixture modelling because it uniquely captures the scenario where data is made up of normally distributed sub populations. We ran the algorithm over all covariance types as defined by the scikit-learn [4] implementation, and number of components from 2 to 20 keeping the best resulting score. Lastly we used DB-SCAN, since this algorithm captures density based clusters. To set the hyper parameter for this algorithm we followed the suggested steps from the authors of DBSCAN [7]. Table 2 reports the silhouette score for all datasets using each method. Similarly, Table 3 reports the Davies-Bouldin index for each clustering method on all datasets.

3.4 ATOMIC's Interpretability

One metric for interpretability is the number of features used in the optimal clustering solution found by ATOMIC. Table 4 presents the number of variables used by ATOMIC in it's best clustering solution, a smaller set indicates greater interpretability. In addition, we present the results of ATOMIC on the ceramic dataset [1] as an example which visualizes the interpretability of ATOMIC. We begin by presenting the clustering solutions found by k-means clustering. Since this dataset has 19 dimensions, we utilize the density persevering visualization method densMAP [16] to project the result into three dimensions, as shown in Figure 1. We need to rely on densMAP to visualize this, hence the lack of defined axes obscures what the separation corresponds to. ATOMIC separates the data only using 3 variables hence we can plot this without the need for densMAP projections. This means we can see the rationale for data separation by looking at the axes of the plot. This is evidently more interpretable as seen in Figure 1.

4 DISCUSSION

4.1 Clustering Quality

From Tables 2 and 3 it is evident that on the datasets tested, ATOMIC generally outperforms the clustering methodologies it was compared to. In terms of the Davies-Bouldin score it scored an average of 0.6229 points lower, where lower is better, than the second best

performing algorithm. Considering the Silhouette metric ATOMIC outperformed other methods on all but two of the used datasets. ATOMIC outperformed by an average of 0.1531 over the next best performing algorithm. As discussed in Section 3.2 the silhouette metric ranges from $[-1, 1]$ and hence an average over performance of 0.1531 is fairly substantial. We observe that performance tends to scale with the percent reduction between the overall number of variables and the number used in ATOMIC's best separation. Table 4 presents the number of variables used in the best clustering solution for each dataset tested, along with the total variables and the percent reduction between the two. We note that ATOMIC outperformed the traditional clustering algorithms on all datasets with a % reduction of over 60%, only performing worse than k-means or GMM when the reduction was below a 60% difference. Additionally the datasets on which ATOMIC performed the best had the largest % reduction. This indicates that ATOMIC performs best when it can find a small set of features which define a clustering solution amongst an overall larger set.

4.2 Interpretability

Our goal for interpretability was answering "why does my data fit into distinct clusters?". Our experiments show that ATOMIC does answer this question by returning a set of cluster defining variables. ATOMIC was able to successfully locate such a set for all tested datasets.

We highlight the benefits of our interpretable approach by considering the goal presented by the authors of the Chemical Composition of Ceramic Samples dataset. They "hope to identify chemical elements that are strongest explanatory variables to classify samples into different cultural eras" [1]. Considering the results of ATOMIC seen in Figure 1 we can clearly state 'MgO', 'CaO' and 'P2O5' uniquely define the different cultural groups found by ATOMIC. ATOMIC found additional possible clustering solutions each of which also have a simple set of defining variables. We compare this to Expanding Explainable k-Means Clustering *exKMC* [9], which as seen in Figure 2 produced a decision tree to explain its clustering assignments. This method also produces a set of variables defining the clusters, as listed on the tree, however the tree also lists the mistakes it makes. This means that the explanation provided by the tree is inaccurate to the true clustering model, this makes the answer to what "chemical elements that are strongest explanatory variables to classify samples into different cultural eras" less certain for exKMC than for ATOMIC.

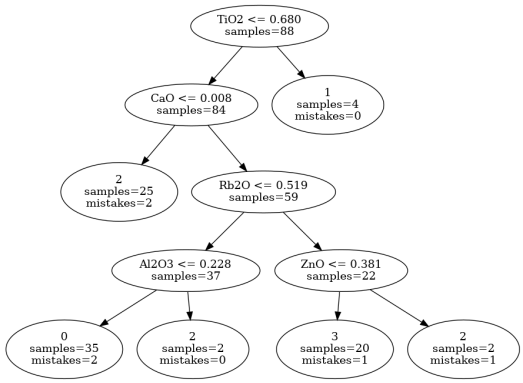


Figure 2: Result of exKMC on Ceramic Dataset

REFERENCES

[1] 2019. Chemical Composition of Ceramic Samples. UCI Machine Learning Repository. DOI: https://doi.org/10.24432/C54P5X.

[2] Sanghamitra Bandyopadhyay and Ujjwal Maulik. 2001. Nonparametric genetic clustering: comparison of validity indices. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)* 31, 1 (2001), 120–125.

[3] Dimitris Bertsimas, Agni Orfanoudaki, and Holly Wiberg. 2021. Interpretable clustering: an optimization approach. *Machine Learning* 110, 1 (01 Jan 2021), 89–138. https://doi.org/10.1007/s10994-020-05896-2

[4] Lars Buitinck, Gilles Louppe, Mathieu Blondel, Fabian Pedregosa, Andreas Mueller, Olivier Grisel, Vlad Niculae, Peter Prettenhofer, Alexandre Gramfort, Jaques Grobler, Robert Layton, Jake Vanderplas, Arnaud Joly, Brian Holt, and Gaël Varoquaux. 2013. API design for machine learning software: experiences from the scikit-learn project. arXiv:1309.0238 [cs.LG]

[5] Arun Das and Paul Rad. 2020. Opportunities and Challenges in Explainable Artificial Intelligence (XAI): A Survey. arXiv:2006.11371 [cs.CV]

[6] Hossein Esfandiari, Vahab Mirrokni, and Shyam Narayanan. [n. d.]. *Almost Tight Approximation Algorithms for Explainable Clustering*. 2641–2663. https://doi.org/10.1137/1.9781611977073.103 arXiv:https://epubs.siam.org/doi/pdf/10.1137/1.9781611977073.103

[7] Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. 1996. A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining* (Portland, Oregon) *(KDD'96)*. AAAI Press, 226–231.

[8] Andrey Filchenkov, Sergey Muravyov, and Vladimir Parfenov. 2016. Towards cluster validity index evaluation and selection. In *2016 IEEE Artificial Intelligence and Natural Language Conference (AINL)*. 1–8.

[9] Nave Frost, Michal Moshkovitz, and Cyrus Rashtchian. 2020. ExKMC: Expanding Explainable k-Means Clustering. arXiv:2006.02399 [cs.LG]

[10] Jacob Kauffmann, Malte Esders, Lukas Ruff, Gregoire Montavon, Wojciech Samek, and Klaus-Robert Muller. 2022. From Clustering to Cluster Explanations via Neural Networks. *IEEE Transactions on Neural Networks and Learning Systems* (2022), 1–15. https://doi.org/10.1109/tnnls.2022.3185901

[11] Markelle Kelly, Rachel Longjohn, and Kolby Nottingham. 2022. *The UCI Machine Learning Repository*. https://archive.ics.uci.edu

[12] Joel Lehman and Kenneth O Stanley. 2011. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary computation* 19, 2 (2011), 189–223.

[13] Joel Lehman and Kenneth O. Stanley. 2011. Evolving a Diversity of Virtual Creatures through Novelty Search and Local Competition. In *Proceedings of the 13th Annual Conference on Genetic and Evolutionary Computation* (Dublin, Ireland) *(GECCO '11)*. Association for Computing Machinery, New York, NY, USA, 211–218. https://doi.org/10.1145/2001576.2001606

[14] W. James Murdoch, Chandan Singh, Karl Kumbier, Reza Abbasi-Asl, and Bin Yu. 2019. Definitions, methods, and applications in interpretable machine learning. *Proceedings of the National Academy of Sciences* 116, 44 (2019), 22071–22080. https://doi.org/10.1073/pnas.1900654116 arXiv:https://www.pnas.org/doi/pdf/10.1073/pnas.1900654116

[15] Gregory Naitzat, Andrey Zhitnikov, and Lek-Heng Lim. 2020. Topology of deep neural networks. arXiv:2004.06093 [cs.LG]

[16] Ashwin Narayan, Bonnie Berger, and Hyunghoon Cho. 2020. Density-Preserving Data Visualization Unveils Dynamic Patterns of Single-Cell Transcriptomic Variability. *bioRxiv* (2020). https://doi.org/10.1101/2020.05.12.077776

[17] Cynthia Rudin. 2019. Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead. arXiv:1811.10154 [stat.ML]

[18] Ketan Rajshekhar Shahapure and Charles Nicholas. 2020. Cluster quality analysis using silhouette score. In *2020 IEEE 7th international conference on data science and advanced analytics (DSAA)*. IEEE, 747–748.

[19] Mihir Shekhar, Lini Thomas, and Kamalakar Karlapalem. 2018. High Dimensional Clustering: A Strongly Connected Component Clustering Solution (SCCC). In *2018 IEEE International Conference on Data Mining Workshops (ICDMW)*. 1104–1111. https://doi.org/10.1109/ICDMW.2018.00159

[20] Haoyu Yang, Lianmeng Jiao, and Quan Pan. 2021. A Survey on Interpretable Clustering. In *2021 40th Chinese Control Conference (CCC)*. 7384–7388. https://doi.org/10.23919/CCC52363.2021.9549986

[21] Ryan Zhou and Ting Hu. 2023. Evolutionary approaches to explainable machine learning. In *Handbook of Evolutionary Machine Learning*. Springer, 487–506.